

AI ASSISTED LAB ASSIGNMENT-6.1

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Task-1: (AI-Based Code Completion for Loops)

Prompt:

“Generate Python code to print all even numbers between 1 and N

Using a loop.”

Input:

#Generate Python code to print all even numbers between 1 and N using a loop.

```
N = int(input("Enter a number: "))
print("Even numbers between 1 and", N, "are:")
for i in range(1, N + 1):
    if i % 2 == 0:
```

```
print(i)
```

output:

```
PS C:\Users\Prasad_reddy\.vscode> & "C:\Users\Prasad_reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad_reddy/.vscode/python.py/lab.py"
Enter a number: 10
Even numbers between 1 and 10 are:
2
4
6
8
10
PS C:\Users\Prasad_reddy\.vscode> 
```

Task-2: (AI-Based Code Completion for Loop withConditionals)

Prompt:

“Generate Python code to count how many numbers in a list are even and odd.”

Input:

```
def count_even_odd(numbers):
```

```
    even_count = 0
```

```
    odd_count = 0
```

```
    for num in numbers:
```

```
        if num % 2 == 0:
```

```
            even_count += 1
```

```
        else:
```

```
            odd_count += 1
```

```
    return even_count, odd_count
```

```
# Example usage
```

```
numbers_list = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
even_count, odd_count = count_even_odd(numbers_list)
print(f"Even numbers count: {even_count}")
print(f"Odd numbers count: {odd_count}")
```

output:

```
10
PS C:\Users\Prasad_reddy\.vscode> & "C:\Users\Prasad_reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad_reddy/.vscode/python.py/lab.py"
Even numbers count: 5
Odd numbers count: 5
PS C:\Users\Prasad_reddy\.vscode>
```

Task-3: (AI-Based Code Completion for Class Attributes Validation)

Prompt:

“Generate a Python class User that validates age and email using conditional statements.”

Input:

```
class User:

    def __init__(self, name, age, email):

        self.name = name

        self.age = age

        self.email = email

    def validate_age(self):

        if self.age < 0:
```

```

        return "Age cannot be negative."

    elif self.age < 18:

        return "User is a minor."

    else:

        return "User is an adult."

def validate_email(self):

    if "@" not in self.email or "." not in self.email:

        return "Invalid email format."

    else:

        return "Email is valid."

# Example usage:

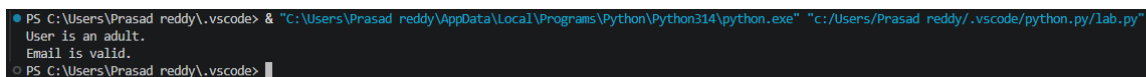
user1 = User("Alice", 25, "alice@example.com")

print(user1.validate_age()) # Output: User is an adult.

print(user1.validate_email()) # Output: Email is valid.

```

Output:



```

PS C:\Users\Prasad_reddy\.vscode> & "C:\Users\Prasad_reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad_reddy/.vscode/python.py/lab.py"
User is an adult.
Email is valid.
PS C:\Users\Prasad_reddy\.vscode>

```

Task-4: (AI-Based Code Completion for Classes)

Prompt:

“Generate a Python class Student with attributes (name, roll number, marks) and methods to calculate total and average marks.”

Input:

```
class Student:

    def __init__(self, name, roll_number, marks):

        self.name = name

        self.roll_number = roll_number

        self.marks = marks


    def calculate_total(self):

        return sum(self.marks)


    def calculate_average(self):

        total = self.calculate_total()

        return total / len(self.marks) if self.marks else 0


# Example usage:

student1 = Student("Bob", 101, [85, 90, 78, 92])

print(f"Total marks for {student1.name}: {student1.calculate_total()}")

print(f"Average marks for {student1.name}: {student1.calculate_average()}")
```

output:

```
PS C:\Users\Prasad_reddy\.vscode> & "C:\Users\Prasad_reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad_reddy/.vscode/python.py/lab.py"
Total marks for Bob: 345
Average marks for Bob: 86.25
PS C:\Users\Prasad_reddy\.vscode> 
```

Task-5: (AI-Assisted Code Completion Review)

Prompt:

“Generate a Python program for a simple bank account system using class, loops, and conditional statements.”

Input:

#Generate a Python program for a simple bank account system using class, loops, and conditional statements.

class BankAccount:

```
def __init__(self, account_number, account_holder, balance=0):  
    self.account_number = account_number  
    self.account_holder = account_holder  
    self.balance = balance
```

```
def deposit(self, amount):
```

```
    if amount > 0:  
        self.balance += amount  
        print(f"Deposited: ${amount}. New balance: ${self.balance}.")  
    else:  
        print("Deposit amount must be positive.")
```

```
def withdraw(self, amount):
```

```
    if amount > 0:  
        if amount <= self.balance:  
            self.balance -= amount  
            print(f"Withdrew: ${amount}. New balance: ${self.balance}.")  
        else:  
            print("Insufficient funds.")  
    else:  
        print("Withdrawal amount must be positive.")
```

```
def check_balance(self):  
    print(f"Account Balance: ${self.balance}.")  
    #give example usage  
if __name__ == "__main__":  
    # Create a new bank account  
    account = BankAccount("12345", "John Doe", 1000)  
  
    # Check initial balance  
    account.check_balance()  
  
    # Deposit money  
    account.deposit(500)  
  
    # Withdraw money  
    account.withdraw(200)  
  
    # Check final balance  
    account.check_balance()
```

Output:

```
PS C:\Users\Prasad reddy\.vscode> & "C:\Users\Prasad reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad reddy/.vscode/python.py/lab.py"  
Account Balance: $1000.  
Deposited: $500. New balance: $1500.  
Withdraw: $200. New balance: $1300.  
Account Balance: $1300.  
PS C:\Users\Prasad reddy\.vscode> |
```